

13.1.1: Fibonacci Sequence using Modules

ALGORITHM

1. **START**

2. **IMPORT** fibonacci_module.

3. **READ** an integer n from user input.

$n > 0$

4. **IF :**

○ **CALL** fibonacci_module.generate_fibonacci_sequence(n) and store the result.

○ **PRINT** the Fibonacci sequence as space-separated numbers.

5. **ELSE:**

○ **PRINT** "Please enter a positive integer".

6. **END**

The screenshot displays the CodeTANTRA IDE interface. On the left, the problem statement for '13.1.1. Fibonacci Sequence using Modules' is visible, detailing the requirements for a Python program that generates the Fibonacci sequence up to a given maximum value. The main program requirements specify importing the 'fibonacci_module', reading an integer 'n' from user input, validating if 'n' is positive, and calling the module function to display the sequence. The input and output formats are also defined. On the right, the code editor shows the implementation of the 'generate_fibonacci_sequence' function in 'main.py'. The function initializes a list 'fib_sequence' and uses a while loop to generate the sequence up to 'max_value'. The test results panel shows that 4 out of 4 shown test cases passed, with the expected output '10' and actual output '10' for the first test case. The terminal and test cases tabs are also visible at the bottom.

```
def generate_fibonacci_sequence(max_value):  
    fib_sequence = []  
    a, b = 0, 1  
    while a <= max_value:  
        fib_sequence.append(a)  
        a, b = b, a + b  
    return fib_sequence
```

Test Results:

- 4 out of 4 shown test case(s) passed
- 4 out of 4 hidden test case(s) passed
- Test case 1: Expected output 10, Actual output 10
- Test case 2: Passed
- Test case 3: Passed

